CO2_AT2

Question 1: Prompt Engineering for Text Summarization

1. Problem Statement

Design an LLM-based system that summarizes long documents into concise, accurate, and easy-to-read summaries while preserving important information.

Complete Solution

1. LLM Architecture

Choose a transformer-based Large Language Model such as:

- GPT-4/5
- Llama 3
- Mistral
- Claude

Architecture:

```
User
  │
  ▼
Document Input
  │
  ▼
Tokenizer
  │
  ▼
Transformer Encoder-Decoder
```

│

▼

Attention Layers

│

▼

Generated Summary

The transformer uses:

- Self Attention
- Multi-head Attention
- Feed Forward Networks
- Positional Encoding

to understand long contexts.

2. Pre-training and Fine-tuning

Pre-training

The LLM is pre-trained on:

- Books
- Research papers
- Wikipedia
- News
- Websites

Objective:

Predict the next word and learn language patterns.

Example:

Input

Artificial Intelligence is...

Output

transforming industries.

Fine-tuning

Fine-tune the model using summarization datasets**.**

Examples:

- CNN DailyMail
- XSum
- PubMed

The model learns:

Document

↓

Summary

Example

Original:

The company announced a new AI model...

Summary:

The company released a new AI model.

3. Transfer Learning

Instead of training from scratch,

Use an already trained LLM.

Benefits

- Faster
- Lower cost
- Better performance

- Less data needed

Flow

Pre-trained GPT

|

▼

Fine-tune on Summarization

|

▼

Domain-specific Summarizer

4. Prompt Engineering

Good prompts improve summary quality.

Example

Poor Prompt

Summarize this.

Better Prompt

Summarize the following article in exactly 100 words.

Highlight only the important points.

Avoid repetition.

Use simple English.

Prompt Template

Role:

You are an expert summarizer.

Task:

Generate a concise summary.

Length:

100 words

Style:

Professional

Output:

Bullet points

5. Zero-shot, One-shot and Few-shot Prompting

Zero-shot

No examples.

Summarize the following document in 5 bullet points.

One-shot

One example provided.

Example

Input:

Machine learning is...

Output:

Machine learning enables computers to learn from data.

Now summarize:

<Document>

Few-shot

Multiple examples.

Example 1

Input...

Output...

Example 2

Input...

Output...

Now summarize:

Few-shot improves consistency.

6. Reasoning-oriented Prompting

Ask the model to identify key ideas before summarizing.

Example

Read the article.

Step 1:

Find main topics.

Step 2:

Remove unnecessary details.

Step 3:

Generate summary.

Benefits

- Better accuracy
- Better factual consistency
- Less missing information

7. Structured Prompting

Provide a fixed structure.

Example

Summarize using**:**

Title

Overview

Key Points

Conclusion

Output

Title

Overview

Key Points

Conclusion

This creates consistent summaries.

8. Function Calling

External functions can improve summarization.

Example

Extract PDF

↓

Split into pages

↓

LLM summarizes each page

↓

Merge summaries

Possible functions

extract_text()

translate()

count_words()

save_summary()

9. LoRA / PEFT

Fine-tuning entire LLM is expensive.

Use

- LoRA
- PEFT

Advantages

- Small GPU
- Faster training
- Lower memory
- Low cost

Train only adapter layers instead of the complete model**.**

10. LLM API Selection

Suitable APIs

API	Advantages
OpenAI GPT	High-quality summaries
Anthropic Claude	Long context
Google Gemini	Multimodal support
Llama API	Open-source

Selection Criteria

- Cost
- Accuracy
- Speed

- Context length
- **Security**

11. Prompt Evaluation

Metrics

- ROUGE
- BLEU
- BERTScore
- Human evaluation

Check

- Accuracy
- Completeness
- Readability
- Hallucination
- Grammar

Example

Input:

1000 words

Generated Summary:

120 words

Human Score:

9/10

12. Security and Edge-case Handling

Security

- Remove sensitive data
- Prevent prompt injection

- Validate input
- Limit document size

Edge Cases

Very long documents

Solution:

Chunking

Document

↓

Chunk 1

Chunk 2

Chunk 3

↓

Summarize each

↓

Final Summary

Other edge cases

- Empty document
- Unsupported language
- Corrupted PDF
- Duplicate text

Final Workflow

User Uploads Document

↓

Text Extraction

↓

Chunking

↓

Prompt Engineering

↓

LLM Summarization

↓

Post-processing

↓

Final Summary

Advantages

- Saves time
- High accuracy
- Easy to read
- Supports long documents
- Low cost with LoRA

Question 2: Structured Output for Enterprise Reports

1. Problem Statement

Design an LLM solution that converts business documents into structured enterprise reports with predefined sections and machine-readable formats (JSON/XML).

Complete Solution

1. LLM Architecture

Architecture

Enterprise Documents

↓

Tokenizer

↓

Transformer LLM

↓

Structured Prompt

↓

JSON Generator

↓

Enterprise Dashboard

The transformer understands business documents and generates structured reports.

2. Pre-training and Fine-tuning

Pre-training

The LLM learns from

- Business reports

- Financial documents
- Company policies
- Annual reports
- Public datasets

Fine-tuning

Fine-tune using enterprise-specific datasets.

Example

Input

Sales increased by 15%.

Expenses decreased by 5%.

Output

JSON

```
{
"sales_growth":"15%",
"expense_reduction":"5%"
}
```

3. Transfer Learning

Use a pre-trained business LLM and adapt it for enterprise reporting instead of training from scratch.

Benefits

- Lower cost
- Faster deployment
- Better accuracy
- Less labeled data

4. Prompt Engineering

Example Prompt

You are a business analyst.

Read the report.

Return only JSON.

Include

Executive Summary

Revenue

Expenses

Risks

Recommendations

No extra explanation.

5. Zero-shot, One-shot and Few-shot Prompting

Zero-shot

Convert this report into JSON.

One-shot

Input:

Profit increased.

Output:

{

"profit":"increased"

}

Now convert this report.

Few-shot

Provide several report-to-JSON examples to improve formatting consistency and accuracy.

6. Reasoning-oriented Prompting

Step 1:

Identify business metrics.

Step 2:

Find risks.

Step 3:

Generate structured report.

Step 4:

Validate output.

This helps ensure that important information is not missed.

7. Structured Prompting

Example

Generate

Executive Summary

Financial Metrics

Operational Metrics

Risks

Recommendations

Output only JSON.

Sample Output

JSON

```
{
  "executive_summary": "...",
  "financial_metrics": {
    "revenue": "₹25 Cr",
```

```
    "profit": "₹4 Cr"
  },
  "operational_metrics": {
    "production": "1200 units/day"
  },
  "risks": [
    "Supply chain delay"
  ],
  "recommendations": [
    "Increase inventory"
  ]
}
```

8. Function Calling

Functions

extract_pdf()

extract_excel()

validate_json()

store_database()

generate_dashboard()

Workflow

Document

↓

Extract Data

↓

LLM

↓

Function validates JSON

↓

Database

↓

Dashboard

9. LoRA / PEFT

Use LoRA or PEFT to adapt the model to:

- Company report formats
- Industry-specific terminology
- Internal reporting styles

Benefits

- Lower training cost
- Faster updates
- Reduced GPU requirements

10. LLM API Selection

API	Advantages
OpenAI GPT	Excellent structured outputs and function calling
Anthropic Claude	Strong long-document processing
Google Gemini	Good multimodal document understanding
Llama API	Self-hosting and customization

Selection factors:

- JSON reliability
- Function-calling support
- Security

- Latency
- Cost
- Context window

11. Prompt Evaluation

Evaluate using:

- JSON schema validation
- Field completeness
- Accuracy of extracted values
- Hallucination rate
- Human review

Example Checklist

- Valid JSON
- Required fields present
- Correct financial values
- No extra text

12. Security and Edge-case Handling

Security

- Mask confidential information (PII)
- Encrypt sensitive reports
- Apply role-based access control
- Validate inputs against prompt injection
- Log API usage securely

Edge Cases

- Missing financial fields
- Corrupted documents

- Multiple currencies
- Ambiguous dates
- Empty reports
- Invalid JSON generation (use schema validation and retries)

Final Workflow

Enterprise Report

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Document Parsing

↓

Prompt Engineering

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LLM Processing

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Structured JSON Generation

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Schema Validation

↓

Database Storage

↓

Analytics Dashboboar

Advantages

- Consistent enterprise reports
- Machine-readable structured output
- Easy integration with ERP/BI systems
- Reduced manual effort

- Secure, scalable, and cost-effective with LoRA/PEFT